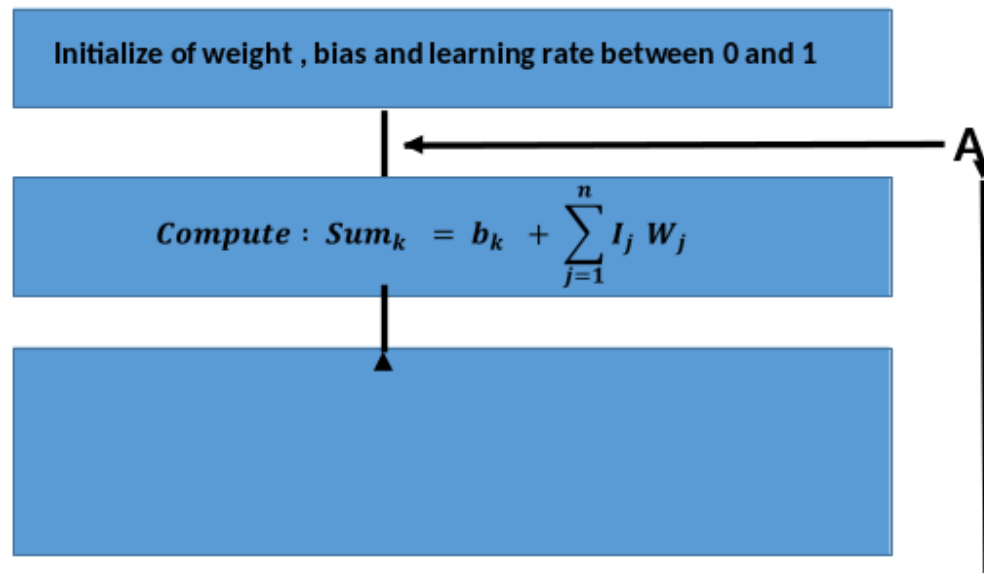


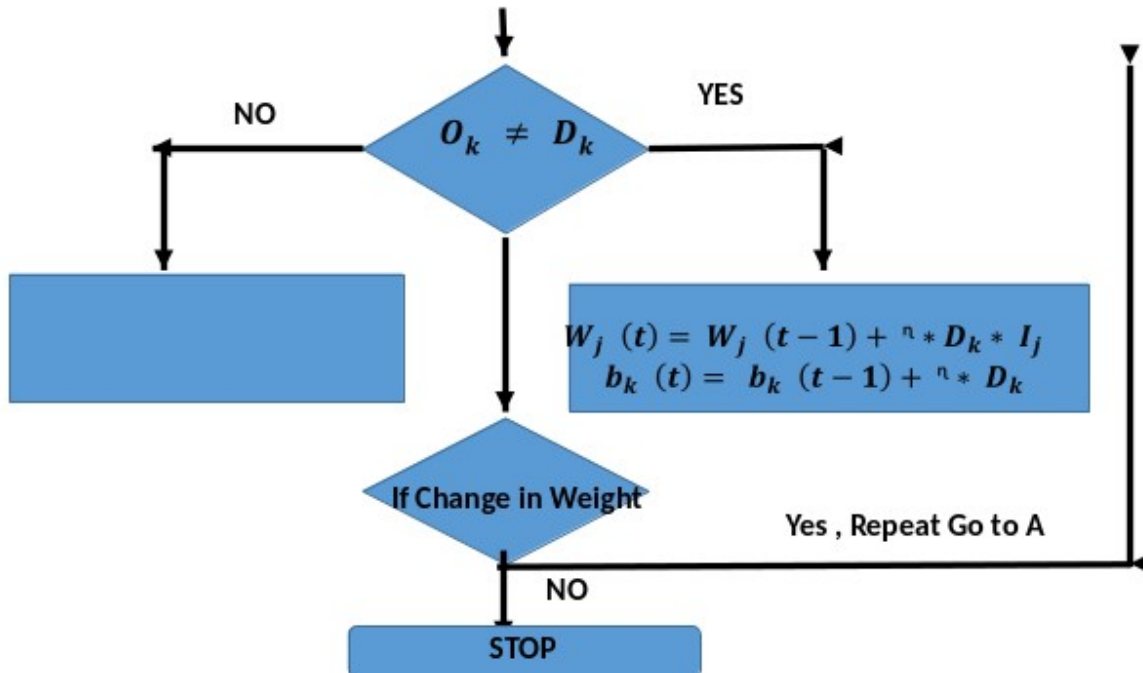
Assignment 5

Q1. Implement the OR and AND problem using perceptron learning model.

Algo

Perceptron Network





- transfer function used in the output neuron of the perceptron

$$f(\text{Sum}_k) = \begin{cases} 1 & \text{if } \text{Sum}_k > \theta_k \\ 0 & \text{if } -\theta_k \leq \text{Sum}_k \leq \theta_k \\ -1 & \text{if } \text{Sum}_k < -\theta_k \end{cases}$$

- $\theta_k = 0.5$, $W_1=0$, $W_2=1$, $n=1$
- Then the weights will be updated

Implement AND and OR.

In each iteration outputs updated weights and t-y and output.

Q2. Implement same with vector notation.

Where $y=g(W^T X)$ and $W^T = W^T(\text{old}) + \alpha(t-y).X^T$

Q3. Download the Fashion MNIST dataset from <https://github.com/zalandoresearch/fashion-mnist> and develop an image classification

